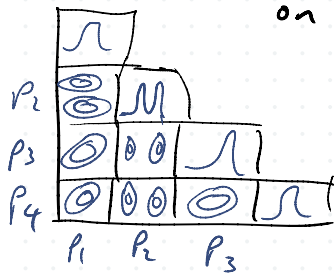


Corner plots



Put all the 2D marginal distributions on the same plot! (And the 1D ones)

They still lose some information, but are a great plotting tool.

EX:

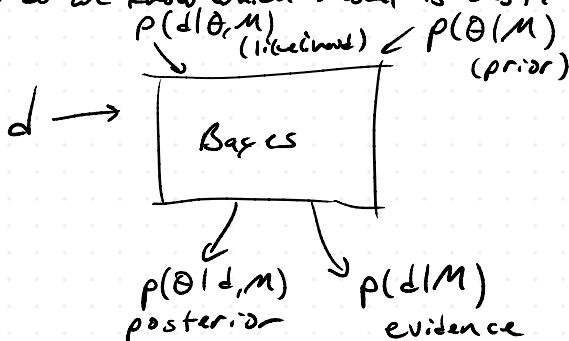


This gap is invisible in all projections

Model Selection

Posterior: prob. of parameters given a model.

How do we know which model is best?



$p(d|M)$: the opposite of what we want!

$\Rightarrow p(M|d)$ Use Bases then!

$$p(d|M) = \frac{p(M|d) p(d)}{p(M)}$$

$$p(M|d) = \frac{p(d|M) p(M)}{p(d)}$$

prior belief in the model

$\int dM p(d|M) p(M)$

The space of all possible models is impracticable

innumerable, or at least impractical to use.

⇒ Compare two models and then $p(d)$ will cancel.
 M_0, M_1

$$\underbrace{\frac{p(M_0|d)}{p(M_1|d)}}_{\text{"Odds ratio" posterior odds}} = \underbrace{\frac{p(d|M_0)}{p(d|M_1)}}_{\text{Evidence ratio "Bayes factor"}} \underbrace{\frac{p(M_0)}{p(M_1)}}_{\text{prior odds (A lot of times, this is 1)}}$$

~~Ex~~

$$M_0: p(\theta) = \delta(\theta)$$

$$M_1: p(\theta) = \mathcal{N}(0, \Sigma^2)$$

$$p(d|\theta) \propto e^{-\frac{1}{2}(\theta - \lambda)^2 / \sigma^2} \quad \sigma: \text{noise stdev.}$$

$$p(\theta|d, M) = \frac{p(d|\theta, M) p(\theta|M)}{p(d|M)}$$

$$\begin{aligned} p(M_0) &\propto \frac{1}{\sigma} \int e^{-\frac{1}{2}(\theta - \lambda)^2 / \sigma^2} \delta(\theta) d\theta \\ &= \frac{1}{\sigma} e^{-\frac{1}{2}\lambda^2 / \sigma^2} \end{aligned}$$

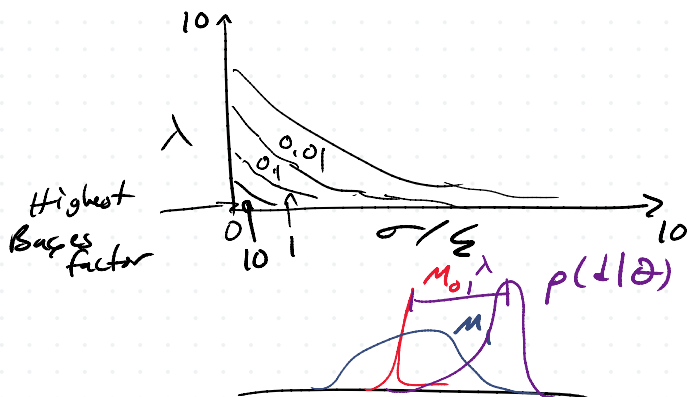
$$p(M_1) \propto \int e^{-\frac{1}{2}(\theta - \lambda)^2 / \sigma^2} e^{-\frac{1}{2}\theta^2 / \Sigma^2} d\theta$$

$$\mathcal{N}(0, \Sigma^2) \cdot \mathcal{N}[\lambda, \sigma^2] = \mathcal{N}(\bar{\mu}, \bar{\sigma}^2)$$

$$\bar{\mu} = \frac{\lambda}{1 + (\sigma/\Sigma)^2}$$

$$\bar{\sigma}^2 = \frac{\sigma^2}{1 + (\sigma/\Sigma)^2}$$

$$\text{Bases factor} = \sqrt{1 + \left(\frac{\epsilon}{\sigma}\right)^2} e^{-\frac{1}{2}\lambda^2 / [1 + \sigma^2/\epsilon^2]}$$



For $\lambda > 1$, M_0 is bad (low Bases factor)
 for $\lambda \sim 1$, it depends on σ/ϵ .

If noise variance is $\gg$ than width of M_1 ,
 then M_1 is good.

For $\sigma/\epsilon \ll 1$: Prior for M_1

$$\log B_{01} \sim \log\left(\frac{\epsilon}{\sigma}\right) - \frac{\lambda^2}{2}$$

How constraining
 is the data?

How informative
 was the data?

$$\log B_{10} \sim \frac{\lambda^2}{2} - \log\left(\frac{\epsilon}{\sigma}\right)$$

Better fit,
 better B_{10}

penalized by
 likelihood to
 prior ratio

More parameters generally give a better fit to
 the data, but constrained parameters penalize
 the Bases factor!

(unconstrained parameters are not penalized)

Bayesian detection statistic:

M_0 : data = noise

M_1 : data = noise + signal

M_2 : data = noise + signal $\rightarrow$ BBH, 11 params

$\rightarrow$ cosmic string,
~5 params

Compare Bayes factors M_2/M_0 , M_1/M_0 , M_2/M_1

What is a "good" Bayes factor?

B_{01}	M_0 or M_1
1	equal
10	prefer M_0 (4/1000 chance wrong)
10^2	
$\vdots$	3σ - 4σ

Bayesian information criteria

#param

$$\text{BIC} = -2 \log(p(d|\theta_{\max})) + k \log N$$

$$\text{AIC} = -2 \log(p(d|\theta_{\max})) + 2k$$

Akaike

$$\text{DIC} = -2 \hat{D}_{\text{KL}} + 2C_N \rightarrow \text{"effective \# of params"}$$

Next class: - approximation methods for evidence and posterior.

$\Rightarrow$ Stochastic sampling algorithms